Empirical Exchange Rate Models of the Seventies: Do They Fit Out of Sample?

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ABSTRACT

Meese and Rogoff ask a narrow but consequential question: once a structural exchange rate model's coefficients are fitted and its explanatory variables are handed their actual realized future values, can it out-predict a naive random walk at horizons of one to twelve months? They test three leading asset-market models of the 1970s — the flexible-price monetary (Frenkel–Bilson), sticky-price monetary (Dornbusch–Frankel), and current-account-augmented (Hooper–Morton) specifications — against the dollar/pound, dollar/mark, dollar/yen, and trade-weighted dollar rates using rolling regressions estimated by ordinary least squares, generalized least squares, and Fair's instrumental-variables method. They also test a battery of univariate time-series models, an unrestricted vector autoregression, and the forward rate. Every alternative, without exception, fails to achieve a lower root mean square forecast error than a driftless random walk. The finding survives price-level substitution, alternative monetary aggregates, alternative inflation proxies, subsample truncation, and combination forecasting. The paper does not explain why the structural models fail; it demonstrates, with unusual rigor for its time, that the failure is real, robust, and specific to out-of-sample evaluation rather than an artifact of in-sample overfitting elsewhere in the literature.

1. Introduction

Richard Meese and Kenneth Rogoff, both economists at the Federal Reserve Board when the paper was written, set out to answer a question that in-sample regression evidence of the 1970s had not addressed: do the era's leading structural exchange rate models actually forecast better than a model that assumes no forecasting is possible at all? The distinction matters because the entire empirical apparatus built up around the monetary approach to exchange rate determination — the flexible-price (Frenkel–Bilson), sticky-price (Dornbusch–Frankel), and current-account-augmented (Hooper–Morton) models — had been validated almost exclusively by how well each model's fitted values tracked the exchange rate over the same sample used to estimate its coefficients. In-sample fit is a weak test of a model's economic content, since a sufficiently flexible specification can track any historical series without describing the mechanism that generated it.

The paper's contribution is not a new theory of the exchange rate. It is a piece of applied econometric discipline: construct the three competing structural models exactly as their proponents specified them, estimate each one repeatedly on expanding samples exactly as a forecaster would have to in real time, hand each model the actual realized values of its own explanatory variables (thereby giving it every benefit of the doubt), and compare the resulting forecast errors against a battery of atheoretical alternatives, foremost among them the random walk. The result — that nothing beats the random walk, at any horizon from one to twelve months, for any of four major dollar exchange rates — became one of the most cited and least comfortable findings in international macroeconomics. It is discussed in this review under the label subsequently attached to it in the literature: the Meese–Rogoff puzzle.

This review reconstructs the paper's nested model specification and forecast-evaluation methodology in full, reproduces its analytically tractable results, and evaluates critically what the out-of-sample failure does and does not establish about exchange rate economics.

2. Research Problem and Hypothesis

2.1 Research Question

The paper asks a precisely bounded question: at forecast horizons of one, three, six, and twelve months, do the flexible-price monetary, sticky-price monetary, and current-account-augmented structural models of the nominal exchange rate — each re-estimated period by period on the most recent available data and each supplied with the true realized values of its own right-hand-side variables — produce lower root mean square forecast errors than a driftless random walk, an estimated univariate time series model, an unrestricted vector autoregression, or the forward exchange rate? The comparison is run separately for the dollar/pound, dollar/mark, dollar/yen, and trade-weighted dollar exchange rates over November 1976 to June 1981.

2.2 Motivation

By the late 1970s a substantial empirical literature reported apparently successful estimates of asset-market exchange rate models: high in-sample R^2, coefficients of roughly the theoretically predicted sign and magnitude, and statistically significant relationships between the exchange rate and relative money supplies, incomes, and interest rates. That literature was used, implicitly or explicitly, to argue that floating exchange rates were governed by identifiable macroeconomic fundamentals and were therefore, in principle, forecastable. The question of whether this apparent success would survive a genuine out-of-sample test had not been systematically posed. It matters because the answer bears directly on whether monetary policy, current-account, and portfolio decisions can be conditioned on model-based exchange rate forecasts, and because it is a direct test of the joint hypothesis that (i) the specified structural relationship is the true data-generating process and (ii) the parameters of that relationship are stable and estimable in real time.

2.3 Existing Literature

The three models under test each descend from the monetary approach to the balance of payments and exchange rate determination. The flexible-price monetary model (Frenkel, 1976; Bilson, 1978, 1979) assumes continuous purchasing power parity and derives the exchange rate from relative money demand. The sticky-price monetary model (Dornbusch, 1976; Frankel, 1979) relaxes continuous PPP, allowing goods prices to adjust slowly while asset markets clear instantaneously, so that a nominal disturbance produces a short-run real exchange rate movement in excess of its long-run equilibrium value — the overshooting mechanism. The Hooper–Morton (1982) model further augments the sticky-price specification with cumulated trade balances as proxies for shifts in the long-run real exchange rate associated with current-account imbalances. Each of these models had been estimated and reported as broadly successful in-sample, principally over the 1973–1978 subperiod of the recent float (Bilson, 1978; Frankel, 1979; Hooper and Morton, 1982).

2.4 Research Gap

What the existing literature had not done, at the time of writing, was hold each model to a forecasting standard that mimics the position of a real-time forecaster: estimate on data available up to date t, forecast forward, and score the forecast against what actually happened, repeating the exercise on a rolling basis as new data arrive. Nor had it benchmarked the structural models against the simplest available null: that today's spot rate is the best available forecast of tomorrow's spot rate. Without that comparison, in-sample coefficient significance cannot be distinguished from overfitting, and the practical forecasting value of the entire monetary-approach research program remains unverified.

2.5 Contribution

PAPER Meese and Rogoff supply exactly that missing comparison, executed with considerable care: rolling out-of-sample re-estimation of three structural models under three different estimators (OLS, GLS, and Fair's instrumental-variables procedure), six univariate time series techniques, an unconstrained six-variable VAR, the forward rate, and the random walk, evaluated by three loss criteria (mean error, mean absolute error, root mean square error) over multiple horizons, multiple currencies, and multiple subperiods.

CRITIQUE The claimed contribution is fully demonstrated for the specific models, sample, and horizon range examined. It is a demonstration, not an explanation: the paper is explicit that it cannot distinguish among several competing accounts of why the structural models fail (Section 5 of the original paper, reconstructed in Section 7 of this review). The contribution is therefore best described as establishing an empirical fact under a stringent test design, not as resolving its cause.

3. Theoretical Framework and Mechanism

The paper is empirical, but every model it tests is a restricted special case of a single reduced-form equation, and that equation itself is not arbitrary: it follows from combining a stable money demand function, purchasing power parity (in either its continuous or its long-run form), and uncovered interest parity. Reconstructing this derivation is necessary to understand what the estimated coefficients mean and why the three competing models are properly described as nested rather than merely similar.

3.1 Background derivation of the nested specification

The following three-equation derivation is reconstructed from the monetary-model literature the paper cites (Frenkel, 1976; Dornbusch, 1976; Bilson, 1978; Frankel, 1979; Hooper and Morton, 1982) rather than re-derived within Meese and Rogoff's own text, which states the reduced form directly in its equation (1) and refers the reader to that literature for its microfoundations.

Let lower-case letters denote logarithms and an asterisk denote the foreign country. Domestic and foreign money demand are each assumed to take the conventional form

$$m - p = \varphi y - \lambda r_s \tag{4}$$

m	log domestic money supply
p	log domestic price level
y	log domestic real income
r_s	domestic short-term nominal interest rate
φ, λ	the income elasticity and interest semi-elasticity of money demand

Equation (4) is reproduced here exactly as it appears in the paper's Section 5, where it is used to motivate the price-level-substitution robustness check; the corresponding foreign equation, $m^*-p^* = \varphi y^*-\lambda r_s^*$, is not separately numbered by the authors but is implied throughout.

Purchasing power parity states that the exchange rate equals the ratio of price levels, $s = p - p^*$. Substituting the two money demand equations to eliminate p and p^* gives an exchange rate determined entirely by relative money supplies, incomes, and interest rates. Uncovered interest parity, $r_s - r_s^* = E[\Delta s]$, closes the model by pinning down the interest differential in terms of expected depreciation. Together these three relationships collapse to the paper's general nested specification:

$$s = a_0 + a_1(m - m^*) + a_2(y - y^*) + a_3(r_s - r_s^*) + a_4(\pi^e - \pi^{e*}) + a_5 \cdot TB + a_6 \cdot TB^* + u \tag{1}$$

s	log of the dollar price of foreign currency
$m-m^*$	log ratio of the U.S. to the foreign money supply
$y-y^*$	log ratio of U.S. to foreign real income
$r_s-r_s^*$	short-term nominal interest rate differential
$\pi^e-\pi^{e*}$	expected long-run inflation differential (unobserved; proxied)
TB, TB^*	cumulated U.S. and foreign trade balances
u	disturbance term, possibly serially correlated

Every coefficient restriction that distinguishes the three candidate models is a restriction on equation (1), which is the paper's central organizing device. All three models impose $a_1=1$ — first-degree homogeneity of the exchange rate in relative money supplies, a direct implication of the underlying money demand equations. The flexible-price Frenkel–Bilson model, which relies on continuous PPP and therefore has no role for sticky prices or current-account wealth effects, additionally sets $a_4=a_5=a_6=0$. The sticky-price Dornbusch–Frankel model allows goods prices to adjust slowly, which reintroduces a role for the expected inflation differential as a proxy for the real interest differential driving temporary deviations from PPP, and sets only $a_5=a_6=0$. The Hooper–Morton model leaves none of the coefficients in equation (1) constrained to zero, adding cumulated trade balances on the premise that unanticipated current-account shocks shift the long-run real exchange rate.

PAPER The economic content of the nesting is therefore a hierarchy of relaxations from strict continuous PPP (Frenkel–Bilson) to short-run nominal rigidity (Dornbusch–Frankel) to current-account-driven long-run real exchange rate shifts (Hooper–Morton). **CRITIQUE** This is an elegant nesting for hypothesis testing in principle, but the paper does not test the restrictions formally within its forecasting exercise; it estimates each restricted model separately and compares forecast errors, which conflates the truth of the nesting restrictions with sampling variability in a way that a Wald or likelihood-ratio test of the restrictions, conducted in-sample, would not.

3.2 Estimation and the endogeneity problem

Ordinary least squares estimation of equation (1) is potentially inconsistent because several right-hand-side variables — relative money supplies, relative incomes, the interest differential — are endogenous in the underlying theoretical models even though they are treated as exogenous regressors in empirical practice. The paper addresses this by also estimating with generalized least squares (correcting for first-order serial correlation, a Cochrane–Orcutt-type correction) and with Fair's (1970) instrumental variables technique, which treats money supplies, short-term interest rates, and expected long-run inflation rates as endogenous. PAPER The authors report that Fair's method is their primary specification and that generalized least squares "did not yield inferior forecasts" despite its weaker consistency properties under endogeneity.

3.3 The vector autoregression as an atheoretical multivariate benchmark

To ensure that the structural models are not being unfairly disadvantaged by their exogeneity assumptions, the paper also estimates an unconstrained vector autoregression over the exchange rate and the explanatory variables of equation (1):

$$s_t = \sum_{i=1}^{n} \alpha_i s_{t-i} + \sum_{j=1}^{n} \beta_j' X_{t-j} + u_t \quad (2)$$

X_{t-j} vector of the explanatory variables in equation (1), lagged j periods

n uniform lag length across the (six) equations, selected by Parzen's (1975) criterion

u_t serially uncorrelated but possibly contemporaneously correlated across equations

Because no variable is restricted to be exogenous, the VAR is immune to the specific endogeneity critique that applies to OLS estimation of equation (1). Its failure to outpredict the random walk is therefore a materially different and, in one sense, more damaging result than the structural models' failure: it rules out the possibility that the structural models are failing merely because of poor treatment of endogenous regressors, since a model that imposes no exogeneity restriction whatsoever performs no better.

3.4 Forecast evaluation criteria

Out-of-sample accuracy is scored by three statistics computed over the rolling forecasts:

$$ME = \sum_{s=0}^{N_k-1} [F(t+s+k) - A(t+s+k)] / N_k \quad (3a)$$

$$MAE = \sum_{s=0}^{N_k-1} |F(t+s+k) - A(t+s+k)| / N_k \quad (3b)$$

$$RMSE = \{ \sum_{s=0}^{N_k-1} [F(t+s+k) - A(t+s+k)]^2 / N_k \}^{1/2} \quad (3c)$$

k forecast horizon (1, 3, 6, or 12 months)

N_k number of forecasts at horizon k in the evaluation period

$F(t), A(t)$ the forecast value and the actual (realized) value at date t

Because the forecasted variable is the logarithm of the exchange rate rather than its level, these statistics are approximately unit-free (interpretable in percentage terms) and comparable across currency pairs, and the paper's use of the log avoids the asymmetry that Jensen's inequality would otherwise introduce between forecasting the dollar/mark rate and forecasting its reciprocal, the mark/dollar rate. Root mean square error is the paper's primary criterion, since it penalizes large errors quadratically; mean absolute error is reported as a robustness check against the possibility that exchange rate innovations follow a fat-tailed, possibly infinite-variance distribution, for which squared-error criteria are not well behaved; mean error checks for systematic directional bias.

3.5 What the theory predicts, and what would falsify it

If any of the three structural models is the true data-generating process and its parameters are stable and consistently estimable, then — supplied with the actual realized values of its explanatory variables, as this experiment supplies them — that model's forecasts should dominate a random walk's at every horizon in a sufficiently large sample, since the random walk uses no information beyond the current spot rate while the structural model uses genuine macroeconomic information plus knowledge of its own future realization. INFERENCE The clean testable implication is therefore: $RMSE_{\text{structural}} < RMSE_{\text{random walk}}$ at every horizon, asymptotically. The paper's entire empirical exercise is a test of this implication against the four exchange rates and the 1976–1981 sample, and, as Section 5 below documents, the implication fails uniformly.

4. Data and Empirical Strategy

4.1 Sample and frequency

All data are monthly, seasonally unadjusted, spanning March 1973 (the start of generalized floating) through June 1981. Each model is first estimated on data through November 1976 and used to generate forecasts at one-, three-, six-, and twelve-month horizons; the sample is then extended one month at a time and every model is re-estimated (a genuine rolling-window design, not a single in-sample fit) before generating the next set of forecasts. The paper also reports results for two alternative starting/ending points — forecasts beginning November 1978, to check sensitivity to the November 1978 shift in U.S. intervention strategy, and forecasts ending November 1980 — without any change in the qualitative conclusion.

4.2 Variables and sources

Spot and forward rates, short- and long-term interest rates, money supplies (M1-B, M2, and the reserve-adjusted monetary base are all tried), consumer prices, industrial production, and trade balances are drawn from the Federal Reserve Board data base, the OECD Main Economic Indicators, Data Resources Inc., and national sources (the Bundesbank Monthly Report, the Financial Times, and the Bank of Japan's Economic Statistics Monthly) for Germany, the United Kingdom, and Japan respectively. The trade-weighted dollar uses fixed 1972–1976 trade-share weights across ten countries. All data are used in seasonally unadjusted form, a deliberate choice: seasonally adjusted series generated with two-sided filters (Census X-11) implicitly use information that would not have been available to a real-time forecaster, and adjusting variables from different national sources by different methods risks distorting estimated structural parameters. Point-sample (not monthly average) exchange rates are used wherever possible, because Working (1960) showed that monthly averages of a series that follows a random walk on a finer time scale will exhibit spurious positive serial correlation.

4.3 Identification: where does the comparison get its force from?

The design's identifying logic is not cross-sectional or instrumental in the usual sense; it is temporal separation between the information used to fit a model and the information used to score it. Because each model is re-estimated using only data available up to the forecast origin, and because forecast accuracy is then evaluated against data realized strictly after that origin, in-sample overfitting cannot mechanically produce a favorable result the way it can when a fitted model is compared to the same data used to fit it. PAPER The one place the design deliberately departs from a literal real-time forecaster's information set is the treatment of the structural models' explanatory variables: the authors supply the actual realized future values of money supplies, incomes, interest rates, and trade balances, rather than forecasts of those variables. This is explicitly meant to give the structural models their best possible chance — it isolates the question of whether the reduced-form relationship in equation (1) has any forecasting content at all from the separate question of whether its right-hand-side variables are themselves forecastable.

4.4 Potential data limitations

CRITIQUE Several features of the data episode are worth flagging independently of the paper's own discussion. The 1976–1981 window is short by the standards of later exchange rate research (roughly 100 monthly observations per currency, fewer once four years are consumed by the initial estimation period), which limits the power of any out-of-sample comparison to reject a false null. The period also spans two oil shocks and a shift in Federal Reserve operating procedure (October 1979), both of which are exactly the kind of structural break under which any fixed-coefficient model, structural or time series, would be expected to perform poorly regardless of whether its underlying economic content is correct. The paper's own subsample checks (starting in November 1978, ending in November 1980) partially address this, but do not span a full monetary regime on either side of the break with adequate degrees of freedom for separate estimation.

5. Results

5.1 The headline finding

Across every specification the authors report, the driftless random walk achieves the lowest or statistically indistinguishable root mean square forecast error at every horizon from one to twelve months, for all four exchange rates. PAPER In the authors' words, "none of the models achieves lower, much less significantly lower, RMSE than the random walk model at any horizon." The structural models, freely estimated by Fair's instrumental variables method, fail despite being handed the actual realized values of their own explanatory variables — a condition designed to remove any excuse related to the unpredictability of the right-hand-side variables themselves. The forward exchange rate, an entirely different kind of forecaster (a market-based rather than model-based one), also fails to beat the random walk except for the dollar/mark rate at the twelve-month horizon.

5.2 Magnitudes

The random walk itself does not forecast well in an absolute sense; it is simply not beaten by the alternatives. At one month, random walk RMSE for the trade-weighted dollar is 1.99 percent and for the dollar/yen rate 3.70 percent; at twelve months these rise to 8.65 percent and 18.3 percent respectively — large forecast errors by any standard, confirming that no model in this comparison, including the winner, actually explains much of the exchange rate's variation. EVIDENCE This matters for interpretation: the paper does not show that exchange rates are forecastable and the random walk is merely the best available forecaster of a predictable process; the random walk itself performs badly in absolute terms, and everything else performs at least as badly or worse.

5.3 Table analysis: the price-level-substituted specification

Table 1 of the original paper (freely estimated coefficients, Fair's method) is not fully legible in the scanned source consulted for this review; its numeric cells could not be reliably extracted, and are therefore not reproduced here in order to avoid presenting unverified figures. Its qualitative pattern — uniform random walk dominance — is stated unambiguously in the authors' own text quoted above and is corroborated by Table 3, reproduced in full below, which repeats the same comparison after substituting domestic and foreign price levels for the monetary variables in equation (1) (a check discussed in Section 6).

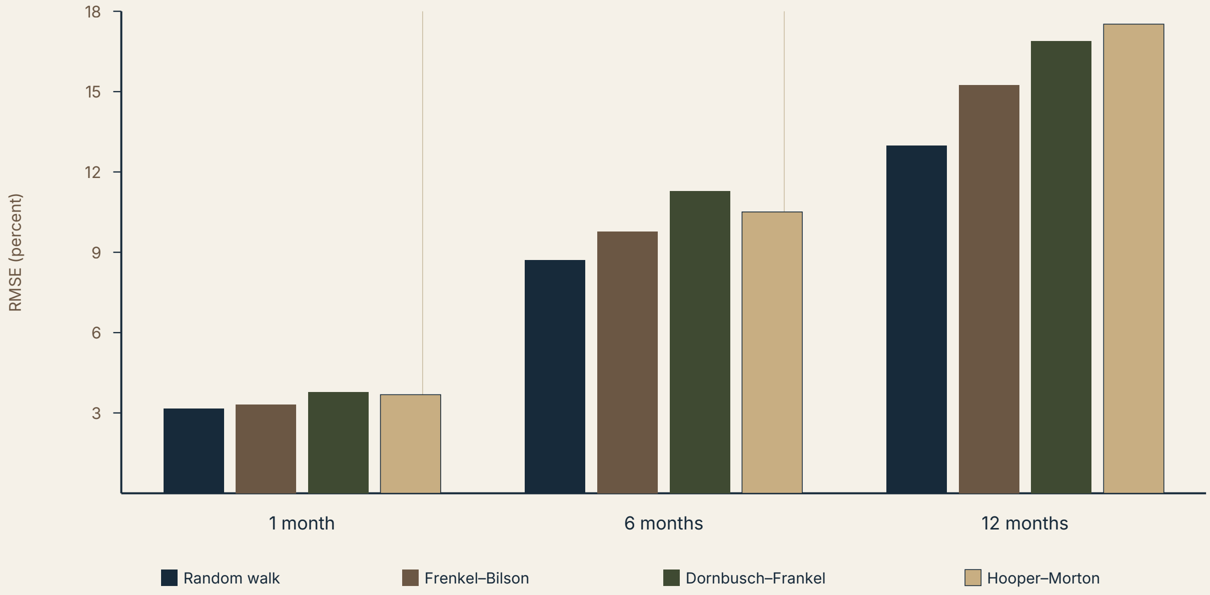
Table 1. **Root mean square forecast errors with price levels substituted for monetary variables (approximately in percentage terms)**

Exchange rate	Horizon	Random walk	Frenkel–Bilson (PPP)	Dornbusch–Frankel	Hooper–Morton
$/mark	1 month	3.17	3.31	3.78	3.68
$/mark	6 months	8.71	9.78	11.28	10.51
$/mark	12 months	12.98	15.25	16.89	17.52
$/yen	1 month	3.70	3.70	4.42	4.18
$/yen	6 months	11.58	12.55	13.86	12.69
$/yen	12 months	18.31	21.80	20.55	20.65
$/pound	1 month	2.56	2.57	2.78	2.94
$/pound	6 months	6.45	7.83	8.64	8.92
$/pound	12 months	9.96	12.51	12.28	14.77
Trade-weighted $	1 month	1.99	2.18	2.29	2.48
Trade-weighted $	6 months	6.09	6.63	6.33	6.80
Trade-weighted $	12 months	8.65	10.48	9.52	9.45

Source: Meese and Rogoff (1983), Table 3, reproduced exactly. Structural models estimated using Fair's instrumental variables technique with a first-order serial correlation correction.

Reading the table by column rather than by row is instructive. In every one of the twelve exchange-rate/horizon combinations, the random walk's RMSE in the left-hand column is the smallest number in its row without exception. The gap widens with the forecast horizon in eight of twelve rows — consistent with errors compounding under a fixed-coefficient structural model when the true process is closer to a random walk than to the model's assumed dynamics — and is largest in absolute terms for the dollar/yen rate at twelve months, where the Hooper–Morton model's RMSE of 20.65 percent exceeds the random walk's 18.31 percent by more than two percentage points.

Figure 1. Out-of-sample RMSE by model and forecast horizon, dollar/Deutsche Mark rate

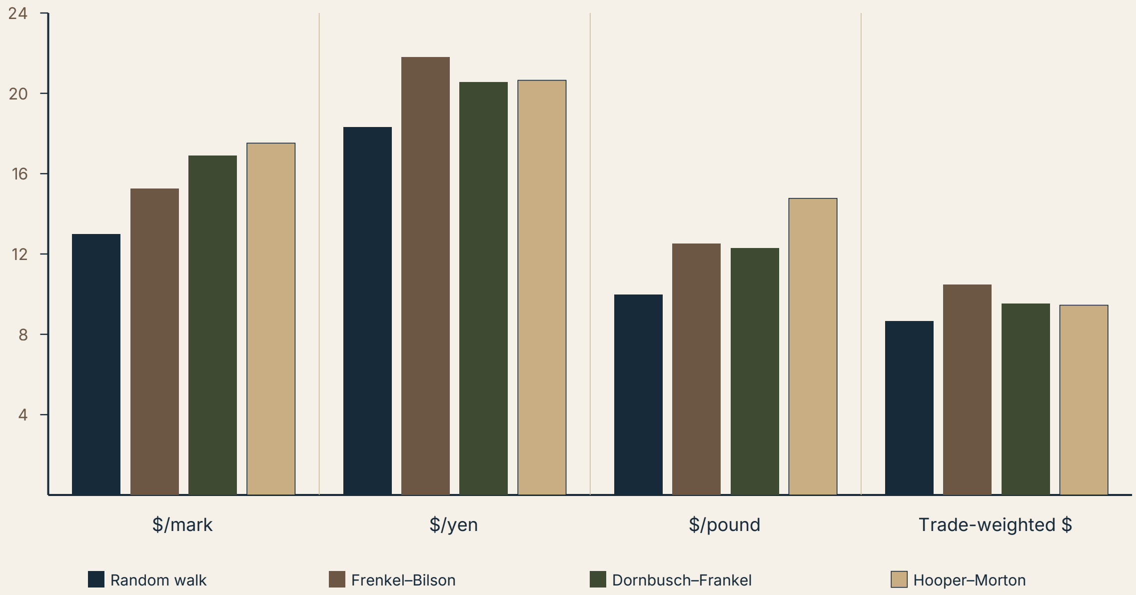


Source: Meese and Rogoff (1983), Table 3 (price-level-substituted specification). Own construction. RMSE of the logarithm of the exchange rate, approximately in percentage terms, November 1976–June 1981.

The chart makes visible what the table states numerically: not only does the random walk (navy) hold the lowest bar in every group, the ranking among the three structural models is unstable across horizons — Frenkel–Bilson is the best-performing structural model at one and six months but the worst at twelve, while Hooper–Morton, which nests the other two and has the most free parameters, is the worst performer at every horizon shown. A model with more free parameters and a more elaborate economic story performing the worst is itself a mild piece of evidence against the "richer model captures more of the true mechanism" interpretation of the nesting in equation (1).

5.4 Cross-currency comparison at the twelve-month horizon

Figure 2. Twelve-month-ahead out-of-sample RMSE, by exchange rate and model



Source: Meese and Rogoff (1983), Table 3, twelve-month rows. Own construction. RMSE of the logarithm of the exchange rate, approximately in percentage terms.

The dollar/yen rate is the least predictable series in the sample by a wide margin at every model, and the trade-weighted dollar — averaging over ten bilateral rates — is the most predictable, consistent with a diversification argument: idiosyncratic bilateral noise partly cancels in a trade-weighted aggregate. The ordering of the four exchange rates by RMSE is the same for every model, which suggests that the pattern reflects the intrinsic volatility of the underlying series rather than any exchange-rate-specific advantage of one model over another.

6. Robustness and Additional Tests

CONCERN → TEST → RESULT

Concern: the structural models might fail only because of the particular monetary aggregate (M1-B) used. **Test:** re-estimate with M2 and with the reserve-adjusted monetary base. **Result:** both alternatives "almost always" yield worse fit and never beat the random walk. **Does it resolve the concern?** Yes, within the range of aggregates tried; it does not rule out that some other unexamined monetary definition would perform differently, but the direction of the evidence (broader/narrower aggregates both worse) argues against aggregate choice being the source of the failure.

CONCERN → TEST → RESULT

Concern: the unobserved expected long-run inflation differential might be poorly proxied by the long-term interest differential used in the baseline specification. **Test:** substitute current-period inflation, a moving average of past inflation, future (realized) inflation, and a twelve-month trailing inflation differential. **Result:** the twelve-month trailing proxy improves the Dornbusch–Frankel and Hooper–Morton dollar/mark forecasts by 0.02 and 0.05 percentage points respectively at one month, and the Hooper–Morton model improves by roughly 2 percent over the random walk for the trade-weighted dollar at twelve months, but every proxy still loses at most horizons and for most currencies. **Does it resolve the concern?** Only partially: the inflation proxy is shown not to be the dominant source of the failure, since even its most favorable specification leaves the random walk ahead in the large majority of horizon/currency cells.

CONCERN → TEST → RESULT

Concern: money demand instability, especially well documented for the United States in this period (Simpson and Porter, 1980), could be corrupting the coefficient estimates through the money-supply terms in equation (1) rather than reflecting a genuine failure of the exchange rate relationship. **Test:** substitute domestic and foreign price levels directly for the monetary variables, using the money demand equation (4) to eliminate money supplies and incomes from the estimating equation, and separately re-estimate the Dornbusch–Frankel model on the three cross-rates (pound/yen, pound/mark, yen/mark) that do not involve the dollar and hence do not involve the particularly unstable U.S. money demand function. **Result:** the price-level-substituted models (Table 3, reproduced in Section 5) still uniformly fail to beat the random walk; the cross-rate re-estimation shows no improvement for pound/yen or pound/mark, but does show the Dornbusch–Frankel model beating the random walk for yen/mark by 0.6 percent at six months and 3.4 percent at twelve months. **Does it resolve the concern?** The isolated yen/mark improvement is suggestive but, in the authors' own assessment, "not so great as to provide a basis for asserting that money demand instability or misspecification is the main problem."

CONCERN → TEST → RESULT

Concern: simultaneous-equations bias from treating endogenous regressors as exogenous could be driving the structural models' poor performance rather than any deficiency in the theory itself. **Test:** estimate with Fair's (1970) instrumental variables procedure treating money supplies, short rates, and expected inflation as endogenous; separately estimate the unconstrained VAR of equation (2), which imposes no exogeneity assumption at all; and, in a companion paper (Meese and Rogoff, 1983, the NBER/University of Chicago Press volume chapter), search a grid of theoretically motivated coefficient constraints rather than freely estimating them. **Result:** none of these approaches produces a structural or multivariate time-series forecaster that beats the random walk at horizons under twelve months. **Does it resolve the concern?** Largely yes — the VAR result in particular is difficult to reconcile with a pure endogeneity story, since it estimates no structural restriction susceptible to endogeneity bias and still loses to the random walk.

CONCERN → TEST → RESULT

Concern: a single point estimate of relative forecasting performance could be an artifact of the particular start and end dates chosen for the evaluation window. **Test:** repeat the entire comparison starting the forecast period in November 1978 (around the shift in U.S. exchange-market intervention policy) and separately ending it in November 1980. **Result:** "the dominance of the random walk model over the other models in RMSE remains" under both alternative windows. **Does it resolve the concern?** Yes, to the extent that the finding is not an artifact of the specific November 1976 starting point, though all three windows remain within the same roughly eight-year floating-rate episode and cannot speak to whether the result would hold in a structurally different monetary regime.

CONCERN → TEST → RESULT

Concern: a combination of several imperfect forecasts might dominate the random walk even if no single forecast does. **Test:** apply Granger and Newbold's (1977) optimal linear combination method, regressing the realized rate on the forecasts of pairs and larger sets of models with weights unconstrained in sign but summing to one. **Result:** a combination of all seven forecasts never beats the random walk; some pairwise combinations do outperform it for a given currency, but "the same combination never works for more than one exchange rate." **Does it resolve the concern?** No stable combination is identified, so this avenue does not overturn the headline result, though the existence of currency-specific pairwise improvements is a genuine, if fragile, qualification worth flagging for future work.

7. Critical Evaluation

7.1 What the paper actually establishes

ESTABLISHED

For the specific models tested (Frenkel–Bilson, Dornbusch–Frankel, Hooper–Morton), the specific currencies (dollar/pound, dollar/mark, dollar/yen, trade-weighted dollar), the specific sample (November 1976–June 1981, with two alternative windows), and horizons of one to twelve months, none of the candidate structural or time-series models achieves a lower root mean square forecast error than a driftless random walk, even when the structural models are supplied with the actual realized values of their own explanatory variables.

SUPPORTED BUT CONDITIONAL

The failure is not attributable to a single choice of monetary aggregate, a single inflation-expectations proxy, a single estimator, or a single sample window — conditional on the specific set of alternatives the authors tried, which is broad but not exhaustive (they did not, for instance, try nonlinear specifications or explicitly model regime shifts around the two oil shocks).

SUGGESTIVE

The pattern of results — more heavily parameterized models (Hooper–Morton, the VAR) performing no better and sometimes worse than more parsimonious ones — is consistent with, though does not prove, an interpretation in which small-sample estimation error in richly parameterized models actively degrades out-of-sample forecast accuracy relative to a parameter-free benchmark.

NOT ESTABLISHED

The paper does not establish why the structural models fail. It explicitly declines to adjudicate among simultaneous-equations bias, sampling error, time-varying true parameters, and misspecification of money demand, purchasing power parity, or uncovered interest parity, stating outright that "determining the relative importance of the possible problems listed above... is at this point speculative." It also does not establish that exchange rates are informationally efficient or that fundamentals are irrelevant to exchange rate determination in any structural sense — only that this particular class of linear, fixed-coefficient reduced-form models cannot exploit those fundamentals for forecasting purposes over this horizon range and sample.

7.2 Strengths

STRENGTHS RECOGNIZED BY THE AUTHORS The paper's own framing emphasizes the methodological contrast with the existing in-sample literature and the robustness of the random walk's dominance across estimators, proxies, and subsamples.

STRENGTHS IDENTIFIED INDEPENDENTLY Three aspects of the design are, in retrospect, unusually disciplined for their time. First, the rolling re-estimation protocol genuinely replicates the information available to a real-time forecaster, which is a stronger design than the split-sample or hold-out procedures common in contemporaneous work. Second, giving the structural models the actual realized values of their explanatory variables is a conservative choice that stacks the deck in the structural models' favor, which makes their failure more rather than less persuasive: a model that cannot forecast even when handed its own future inputs has a specification problem, not merely an input-forecasting problem. Third, the inclusion of the unrestricted VAR as a benchmark is an important control that isolates the finding from the specific criticism that the structural models' exogeneity assumptions, rather than their economic content, are responsible for the failure.

7.3 Weaknesses

WEAKNESS 1 — SMALL SAMPLE, LOW POWER

Problem: the evaluation period contains at most roughly 55 non-overlapping one-month forecasts and correspondingly fewer at longer horizons, drawn from a single historical episode. **Why it matters:** a test with this little independent variation has limited power to distinguish "the structural model is false" from "the structural model is true but sampling error swamped its advantage in this particular five-year window." **Potential bias:** toward failing to reject the random walk even when a true, but modestly informative, structural relationship exists. **How it could be tested:** extending the sample to later decades and additional currency pairs, exactly what the follow-up literature discussed in Section 9 subsequently did.

WEAKNESS 2 — NO FORMAL SIGNIFICANCE TEST OF THE RMSE DIFFERENCES

Problem: the paper compares point estimates of RMSE without a formal statistical test of whether the differences are significant, noting that the available test (Granger and Newbold, 1977) requires independent, normally distributed forecast errors with constant variance and is only strictly applicable at non-overlapping horizons. **Why it matters:** "the random walk has the lowest RMSE in every cell" is a weaker claim than "the random walk is statistically significantly better," and the true economic message of the paper rests on an accumulation of consistent point estimates rather than on any single hypothesis test. **Potential bias:** none in a particular direction, but it leaves open exactly how much confidence the reader should place in any single cell of Table 3 considered alone. **How it could be tested:** the Diebold–Mariano (1995) test, developed roughly a decade later specifically to address this gap, and now the standard tool for exactly this class of comparison.

> **WEAKNESS 3 — REGIME INSTABILITY IS DIAGNOSED BUT NOT MODELED**
>
> **Problem:** the sample spans the second oil shock and the October 1979 shift in Federal Reserve operating procedure, both of which plausibly induce genuine parameter instability in any linear structural model; the paper acknowledges this candidate explanation but does not implement a formal time-varying-parameter estimator (Kalman filtering is mentioned only as a suggestion for future work). **Why it matters:** without directly modeling parameter drift, the paper cannot cleanly separate "the monetary model of exchange rates is fundamentally wrong" from "the monetary model is approximately right but its coefficients shifted twice within a five-year sample too short to re-estimate them separately on either side of each break." **Potential bias:** this could make the structural models look worse than a version with time-varying coefficients would. **How it could be tested:** exactly the class of question later pursued by the state-space and Markov-switching exchange rate literature (see Section 9).

This review considers the single most consequential weakness to be the second: the absence of a formal test of the RMSE differences. It does not overturn the paper's finding — the consistency of the pattern across twelve exchange-rate/horizon cells, multiple estimators, multiple proxies, and multiple sample windows is itself a form of evidence that does not depend on any one significance test — but it means the precise cell-by-cell magnitudes in Table 3 should be read as suggestive of the ranking rather than as individually statistically certified differences.

8. Contradictions and Edge Cases

Under what conditions should the random walk's dominance disappear? The theoretical mechanism underlying equation (1) predicts that a consistently estimated true structural model must eventually outpredict a random walk in a sufficiently large sample, since the random walk discards genuine information that the structural model, by assumption, uses correctly. The paper's own five-year sample may simply be too short for this asymptotic advantage to manifest; a sample spanning several decades and multiple currencies, of the kind assembled by later authors, is the natural test of whether the dominance is a small-sample phenomenon or a persistent one.

What happens at extreme values or under different exchange-rate regimes? The entire exercise is conducted within the recent floating-rate period; it says nothing about fixed or heavily managed exchange-rate regimes, where a monetary or current-account fundamental could in principle be far more informative because the exchange rate itself is not free to absorb news through instantaneous asset-market repricing. Extending the test to a managed-float or target-zone regime is a natural edge-case check, and the answer is not obvious ex ante.

Could another mechanism generate the same observed pattern? A rational, efficient asset market in which the exchange rate is a discounted present value of expected future fundamentals would also generate a series that is very close to a random walk whenever the discount factor applied to those fundamentals is close to one — a mechanism formalized only much later by Engel and West (2005, discussed in Section 9). Under that account, the random walk's dominance in this paper is not evidence against fundamentals mattering for exchange rate determination; it is close to what efficient-markets present-value pricing of a persistent fundamental predicts. Meese and Rogoff's design cannot distinguish "fundamentals are irrelevant" from "fundamentals matter but are already priced in, so the residual is unpredictable," and the paper does not claim to make this distinction.

Does the mechanism produce unreasonable predictions in some regions? Equation (1) imposes identical structural coefficients on domestic and foreign money demand and price-adjustment parameters (footnote 4 of the original paper flags this as a parsimony assumption, following Haynes and Stone, 1981). At the extreme where domestic and foreign economies differ sharply in the interest-elasticity or income-elasticity of money demand — plausible for, say, a large economy paired with a small open economy — this symmetry assumption is a source of misspecification the paper acknowledges but does not relax; the authors report only that "no gain in out-of-sample fit results" from allowing separate coefficients, without exploring more sharply asymmetric parameterizations.

Does the empirical relationship survive different samples? Within the paper, yes: the November 1978 and November 1980 alternative windows do not overturn the finding. Outside the paper, the follow-up literature summarized in Section 9 finds a more nuanced picture — the puzzle is robust across many further decades and currencies at short horizons, but there is more contested evidence of structural-model outperformance at very long horizons (three to five years) and using cointegration- or panel-based techniques not available in 1983.

9. Further Literature

THEORETICAL FOUNDATIONS PRECEDING THE PAPER

Dornbusch (1976), "Expectations and Exchange Rate Dynamics," Journal of Political Economy. Shared idea: the sticky-price overshooting mechanism that generates the Dornbusch–Frankel model tested here. Relationship: Dornbusch supplies the theoretical microfoundation; Meese and Rogoff supply the empirical stress test of its reduced form. Why it matters: without Dornbusch's mechanism there is no economic rationale for the expected-inflation term (a_4) that distinguishes the sticky-price model from the flexible-price Frenkel–Bilson model in equation (1).

Frenkel (1976) and Bilson (1978, 1979), the flexible-price monetary model. Shared idea: continuous purchasing power parity combined with stable money demand. Relationship: the most restricted special case nested inside equation (1). Why it matters: it is the baseline against which the sticky-price and current-account extensions are meant to represent genuine theoretical improvements, yet none of the three outperforms the others out-of-sample.

Hooper and Morton (1982), "Fluctuations in the Dollar," Journal of International Money and Finance. Shared idea: current-account-driven shifts in the long-run real exchange rate. Relationship: the least restricted of the three nested models tested. Why it matters: its poor relative performance in Figure 1 (worst of the three structural models at every horizon shown) is a specific piece of evidence against the view that adding more theoretically motivated variables mechanically improves forecasting performance.

DIRECT EXTENSIONS AND THE NAMING OF THE PUZZLE

Meese and Rogoff (1983), "The Out-of-Sample Failure of Empirical Exchange Rate Models: Sampling Error or Misspecification?," in Frenkel (ed.), Exchange Rates and International Macroeconomics, University of Chicago Press. Shared idea: the same forecasting comparison, extended with a grid search over theoretically constrained coefficients rather than free estimation. Relationship: a companion piece by the same authors, repeatedly cited within the paper reviewed here as the source of the constrained-coefficient robustness results discussed in Section 6. Why it matters: it substantially strengthens the case that simultaneous-equations bias is not the primary explanation, since imposing theoretically sensible coefficient values directly, rather than estimating them, does not rescue the structural models.

Cheung, Chinn, and Pascual (2005), "Empirical Exchange Rate Models of the Nineties: Are Any Fit to Survive?," Journal of International Money and Finance. Shared idea: re-runs the identical logical exercise with 1990s data and an expanded model set including productivity-based and behavioral-equilibrium exchange rate models. Relationship: a direct extension using a different two-decade-later sample. Why it matters: it finds that the original random-walk dominance largely persists, extending the external validity of the 1983 finding well beyond its original sample, though with some evidence of forecast improvement at very long horizons for certain models and currencies.

Cheung, Chinn, Pascual, and Zhang (2019), "Exchange Rate Prediction Redux: New Models, New Data, New Currencies," Journal of International Money and Finance. Shared idea: the same forecasting-tournament design, now including Taylor-rule fundamentals, yield-curve factors, and a wider currency panel. Relationship: the most recent large-scale replication in the same tradition. Why it matters: it reports mixed but still broadly discouraging results for structural models relative to the random walk, indicating that almost four decades of additional data, new fundamentals, and new estimation techniques have not decisively overturned the original puzzle.

PAPERS OFFERING COMPETING EXPLANATIONS

Engel and West (2005), "Exchange Rates and Fundamentals," Journal of Political Economy. Shared idea: reconciling near-random-walk behavior with a genuine role for fundamentals. Relationship: theoretically contradicts the informal reading of Meese–Rogoff as evidence that "fundamentals do not matter," by showing that a rational present-value asset-pricing model with a discount factor close to one generates exchange rate behavior observationally close to a random walk even when fundamentals are the true driving force. Why it matters: it reframes the entire puzzle as a statement about the difficulty of detecting a discounted, forward-looking relationship with short samples, rather than as a refutation of fundamentals-based exchange rate theory.

Evans and Lyons (2002), "Order Flow and Exchange Rate Dynamics," Journal of Political Economy. Shared idea: exchange rate forecasting using an alternative class of explanatory variable. Relationship: contradicts the implicit presumption that no variable can improve on the random walk, by showing that microstructure order flow — net buying pressure observed by market participants but not by the macro econometrician — achieves substantially higher in-sample and some out-of-sample explanatory power at very short (daily) horizons. Why it matters: it relocates the search for exchange rate predictability away from macro fundamentals and toward the trading process itself, a research direction unavailable to Meese and Rogoff in 1983 given data limitations.

Rogoff (1996), "The Purchasing Power Parity Puzzle," Journal of Economic Literature. Shared idea: real and nominal exchange rates are far more volatile and far more persistent in their deviations from fundamentals-implied levels than simple sticky-price models predict. Relationship: a later synthesis, by one of the present paper's own authors, of the broader empirical failure of PPP-based exchange rate theory that this paper's Frenkel–Bilson and Dornbusch–Frankel models both ultimately rest on. Why it matters: it suggests the 1983 out-of-sample failure and the PPP puzzle are two symptoms of the same underlying problem — goods and asset markets do not equilibrate on the time scale these models assume.

10. Research Gaps

PAPER-LEVEL GAPS

The paper does not adjudicate among its own candidate explanations for the structural models' failure (simultaneous-equations bias, sampling error, parameter instability, and misspecification of uncovered interest parity, inflation expectations, goods-market adjustment, or money demand). It also does not test any nonlinear specification, any regime-switching or time-varying-parameter model, or any specification incorporating risk premia directly as an estimated (rather than residual) component.

LITERATURE-LEVEL GAPS

Four decades of follow-up research (Section 9) have established that the puzzle is highly robust at short horizons across samples and currencies, but the literature remains genuinely divided on long-horizon (three-to-five-year) predictability, on whether panel and cointegration techniques applied to fundamentals meaningfully improve on the random walk, and on how to reconcile microstructure-based short-horizon predictability (Evans and Lyons) with the near-total absence of macro-fundamental-based predictability at the same horizons.

METHODOLOGICAL GAPS

Where better data, identification, estimation, or measurement could improve the research: (i) formal out-of-sample significance testing (the Diebold–Mariano test, unavailable in 1983) applied directly to this paper's own currency-horizon cells; (ii) explicit modeling of the 1979 Federal Reserve operating-procedure shift as a structural break rather than as an informal robustness check; (iii) direct estimation of a time-varying risk premium rather than treating deviations from uncovered interest parity as part of the residual; (iv) machine-learning and nonlinear function approximation applied to the same variable set, to test whether the failure is a failure of linearity rather than of the variables themselves.

Gap → Why it matters → Existing limitation → Possible way forward

The clearest remaining gap is methodological: no version of this comparison, from 1983 through the most recent replications, has combined a fully specified structural break model, formal significance testing of RMSE differences, and a genuinely out-of-sample (not just out-of-training-window) design in one exercise. Doing so would finally separate "the monetary approach is wrong" from "the monetary approach is right but was tested across two undetected regime changes with too few degrees of freedom to notice."

11. Research Opportunities

Idea 1 — Diebold–Mariano re-evaluation of the original Meese–Rogoff comparison

RESEARCH QUESTION: Are the RMSE differences reported in Meese and Rogoff (1983) and its 2005/2019 successors statistically significant once tested with a Diebold–Mariano or Clark–West test appropriate for nested model comparisons?

MOTIVATION: the original paper explicitly could not perform this test; it is now standard and could either substantially strengthen or meaningfully qualify seventy years of citations to an unsignificance-tested ranking.

HYPOTHESIS: the random walk's advantage is statistically significant at short horizons (one to three months) but not distinguishable from noise at twelve months, where sample sizes are smallest.

ECONOMIC MECHANISM: forecast error variance from parameter estimation uncertainty grows with horizon under a fixed-coefficient structural model, potentially swamping any true predictive content at longer horizons.

LITERATURE CONNECTION: Diebold and Mariano (1995); Clark and West (2006) for nested-model comparisons; the methodological gap identified in Section 10 above.

DATA: replicate using the same public monthly series (Federal Reserve, OECD Main Economic Indicators) for 1973–1981, plus the extended 1980s–2020s panel from Cheung, Chinn, Pascual, and Zhang (2019).

VARIABLES: identical to equation (1); no new variables required.

METHOD: rolling out-of-sample forecasting exactly as in the original design, with Clark–West statistics computed on each currency-horizon cell.

IDENTIFICATION STRATEGY: temporal train/test separation, as in the original.

EXPECTED CONTRIBUTION: a rigorous statistical verdict on a widely cited but never formally tested ranking.

MAIN RISK: low power at twelve-month horizons given the small effective sample size may make the test uninformative regardless of the true state of the world.

Idea 2 — Explicit structural-break model around the two exchange-rate-relevant regime shifts

RESEARCH QUESTION: does allowing for a discrete break in equation (1)'s coefficients at the October 1979 Federal Reserve operating-procedure change materially improve out-of-sample forecasting relative to the random walk?

MOTIVATION: the original paper flags regime instability as a candidate explanation but tests it only informally via subsample windows.

HYPOTHESIS: a Bai–Perron or Markov-switching version of the Dornbusch–Frankel model outperforms the random walk within each identified regime, even though the pooled, single-regime version does not.

ECONOMIC MECHANISM: if the true money demand or price-adjustment parameters shifted discretely with the change in Federal Reserve operating procedure, a model estimated across the break necessarily averages two different true relationships, degrading its forecasts even if each regime-specific relationship is individually informative.

LITERATURE CONNECTION: Bai and Perron (1998) structural break tests; the regime-shift gap identified in Section 10 above.

DATA: the same 1973–1981 window, supplemented with 1981–1985 data to give the post-break regime adequate degrees of freedom.

VARIABLES: equation (1)'s variable set, plus a regime indicator.

METHOD: Bai–Perron break detection followed by regime-specific rolling re-estimation and forecasting.

IDENTIFICATION STRATEGY: the 1979 policy change is a known, dated, exogenous shift in the monetary policy operating procedure, providing a natural candidate break date rather than one selected by fitting.

EXPECTED CONTRIBUTION: would directly test whether parameter instability, rather than a fundamental absence of forecasting content, explains the original result.

MAIN RISK: the pre- and post-break subsamples individually may be too short for reliable regime-specific estimation, reproducing the small-sample problem at a smaller scale.

Idea 3 — Reconciling order-flow predictability with fundamentals-based unpredictability

RESEARCH QUESTION: can a hybrid model combining Evans and Lyons' (2002) order-flow variable with the Hooper–Morton fundamentals of equation (1) outperform the random walk at horizons between one week and three months, a range intermediate between the daily horizon at which order flow is known to help and the one-to-twelve-month range Meese and Rogoff examine?

MOTIVATION: order flow and macro fundamentals operate on plausibly different time scales; no published comparison spans exactly the intermediate horizon range where their relative contributions are least understood.

HYPOTHESIS: order flow's forecasting advantage decays toward zero by the one-month horizon, while fundamentals contribute nothing incrementally at any horizon tested, so a hybrid model converges to the random walk exactly where Meese and Rogoff found it.

ECONOMIC MECHANISM: order flow aggregates dispersed private information into price relatively quickly; once that information is impounded, only genuinely new macro news should move the rate, and such news is by construction unforecastable.

LITERATURE CONNECTION: Evans and Lyons (2002); the microstructure/macro reconciliation gap identified in Section 10 above.

DATA: proprietary or vendor-supplied daily interbank order-flow data (e.g., EBS or Reuters D2000-2 feeds used in the microstructure literature), aggregated to weekly and monthly frequency, combined with the same macro series as equation (1).

VARIABLES: signed order flow, equation (1)'s fundamentals.

METHOD: rolling out-of-sample forecasting, RMSE and Clark–West comparison against the random walk at weekly, monthly, and quarterly horizons.

IDENTIFICATION STRATEGY: temporal train/test separation; order flow measured strictly prior to the forecast origin.

EXPECTED CONTRIBUTION: would map out exactly where on the horizon spectrum the Meese–Rogoff puzzle begins to bind.

MAIN RISK: proprietary order-flow data are expensive and available for a shorter historical span than the macro series, limiting sample size.

Idea 4 — Nonlinear and machine-learning re-estimation of the identical variable set

RESEARCH QUESTION: does allowing the mapping from equation (1)'s variables to the exchange rate to be nonlinear (via a random forest or a shallow neural network, estimated with the same rolling out-of-sample protocol) recover forecasting content that a linear specification cannot?

MOTIVATION: the original paper tests only linear specifications; if the true relationship involves threshold effects (for example, PPP deviations that only trigger a corrective exchange rate movement once they exceed a transaction-cost band), a linear model would systematically miss it.

HYPOTHESIS: nonlinear models modestly outperform the random walk at long horizons (six to twelve months) where threshold-type PPP reversion is most likely to matter, but not at one month.

ECONOMIC MECHANISM: transaction costs and limits to arbitrage can generate a band of inaction around PPP, inducing nonlinear mean reversion of the type documented in the threshold-autoregression PPP literature.

LITERATURE CONNECTION: threshold autoregression models of real exchange rates (e.g., Taylor, Peel, and Sarno, 2001); the nonlinearity gap noted in Section 10 above.

DATA: the extended Cheung-Chinn-Pascual-Zhang (2019) panel, to obtain enough observations for a nonparametric method.

VARIABLES: equation (1)'s fundamentals, plus their lagged values and simple transformations (deviations from a rolling PPP benchmark).

METHOD: random forest or gradient-boosted trees, trained on expanding windows exactly as in the original rolling design, to preserve genuine out-of-sample evaluation.

IDENTIFICATION STRATEGY: strict temporal separation between training and evaluation folds, with no look-ahead in feature construction.

EXPECTED CONTRIBUTION: would test whether the Meese–Rogoff puzzle is a puzzle of functional form rather than of variable choice.

MAIN RISK: nonlinear methods are prone to in-sample overfitting that masquerades as forecasting skill unless the out-of-sample discipline is enforced exactly as rigorously as in the original paper; a sloppy replication would simply reproduce a subtler version of the very problem Meese and Rogoff set out to avoid.

Idea 5 — Extending the comparison to emerging-market and managed-float currencies

RESEARCH QUESTION: does the random walk's dominance over structural fundamentals models extend to currencies under managed floats or with active central bank intervention, where the exchange rate is not a purely market-cleared asset price?

MOTIVATION: the original paper and nearly all its major successors study only the major freely floating dollar rates; managed and emerging-market currencies, where fundamentals plausibly play a larger role because the exchange rate does not instantaneously absorb news through unrestricted asset trading, remain comparatively unexamined by this specific forecasting-tournament design.

HYPOTHESIS: Hooper–Morton-type current-account-augmented models outperform the random walk more often for managed-float currencies with binding current-account constraints (for example, several emerging-market currencies with capital controls) than for the major floating rates studied here.

ECONOMIC MECHANISM: central bank intervention that targets a current-account or reserve objective directly links the exchange rate's path to the same trade-balance variables that enter the Hooper–Morton specification, potentially restoring genuine forecasting content that pure asset-market pricing eliminates in freely floating rates.

LITERATURE CONNECTION: the broader managed-float and "fear of floating" literature (Calvo and Reinhart, 2002); Mendonça (2025, Journal of Forecasting) on fundamentals versus the random walk in an emerging-market context.

DATA: IMF International Financial Statistics and national central bank data for a panel of managed-float emerging-market currencies over the post-2000 period.

VARIABLES: equation (1)'s fundamentals plus a measure of capital account openness or intervention intensity.

METHOD: the identical rolling out-of-sample RMSE comparison, applied currency by currency and pooled in a panel specification.

IDENTIFICATION STRATEGY: cross-currency variation in intervention intensity used to sort currencies into "more managed" and "more freely floating" subsamples for comparison.

EXPECTED CONTRIBUTION: would test the external validity of the Meese–Rogoff puzzle outside the specific institutional setting (freely floating major currencies) in which it was discovered.

MAIN RISK: data quality and length for emerging-market series are typically weaker and shorter than for the G-7 currencies, which could reproduce the original paper's own small-sample power problem in a more acute form.

11.1 Ranking

Ranked by a combination of novelty, feasibility, data availability, methodological quality, and potential contribution, the strongest three ideas are, in order: Idea 1 (Diebold–Mariano re-evaluation), Idea 2 (structural-break re-estimation), and Idea 4 (nonlinear re-estimation).

Idea 1 ranks first because it is the most feasible — it requires no new data beyond what the original and follow-up papers already used — and because it closes a genuine, well-defined methodological gap (the absence of formal significance testing) that has been noted in this review as the single most important weakness of the original design. Idea 2 ranks second: it is directly motivated by an explanation the original authors themselves proposed but did not formally test, has a natural, exogenously dated break point, and would meaningfully advance understanding of whether the puzzle reflects parameter instability rather than a deeper absence of forecasting content. Idea 4 ranks third: it is the most novel of the five and addresses a real gap (linearity), but carries the highest methodological risk, since a nonlinear model that is not rigorously held to the same rolling out-of-sample discipline as the original paper could manufacture the appearance of forecasting skill through subtler overfitting. Ideas 3 and 5 are promising but rank lower principally on data feasibility grounds: proprietary order-flow data and long, high-quality emerging-market series are each harder to obtain than the inputs required for Ideas 1, 2, and 4.

12. Learning Framework

PREREQUISITE CONCEPTS

Uncovered interest parity; purchasing power parity (absolute and relative, continuous and long-run); the money demand function and its income and interest elasticities; the mechanics of nested hypothesis testing via coefficient restrictions; ordinary least squares versus generalized least squares versus instrumental variables under endogenous regressors; the concept and construction of a vector autoregression; root mean square error, mean absolute error, and mean error as forecast-loss criteria; the distinction between in-sample fit and genuine out-of-sample forecasting.

CORE CONCEPTS LEARNED

That in-sample statistical significance of a macroeconomic relationship is not evidence that the relationship has forecasting value; that a rolling, real-time re-estimation design is the appropriate way to test forecasting claims; that giving a model the benefit of the doubt on its explanatory variables' future values isolates the forecasting content of the estimated relationship itself from the separate problem of forecasting the inputs; and that a battery of robustness checks across estimators, proxies, and subsamples is what converts a single suggestive result into a durable stylized fact.

IMPORTANT EQUATIONS

The nested reduced-form specification, equation (1), and its three restricted special cases; the money demand equation (4) used in the price-level-substitution robustness check; the vector autoregression, equation (2), as an atheoretical multivariate benchmark; and the three forecast-loss statistics, equations (3a)–(3c).

METHODS LEARNED

Rolling-window out-of-sample forecast evaluation; Fair's (1970) instrumental variables estimator for models with lagged endogenous variables and serially correlated errors; vector autoregression as a robustness benchmark against exogeneity assumptions; the Granger-Newbold (1977) method for combining forecasts; and the general logic of nested model comparison via coefficient restrictions.

QUESTIONS TO TEST UNDERSTANDING

Comprehension. Can you state, without looking back at the paper, exactly what distinguishes the Frenkel–Bilson, Dornbusch–Frankel, and Hooper–Morton models as restrictions on equation (1), and why each restriction corresponds to a specific economic assumption?

Analysis. Why does the paper's decision to supply the structural models with actual realized values of their explanatory variables make the finding harder, rather than easier, to dismiss? Why is the failure of the unconstrained VAR a more damaging result for the endogeneity critique than the failure of the OLS-estimated structural models alone?

Research. If you were to re-run this exact comparison today with forty additional years of data, what is the single design choice you would change first, and why — and what result would most surprise you?

PAPERS TO READ NEXT

Dornbusch (1976), for the theoretical mechanism this paper's central sticky-price model rests on; Cheung, Chinn, and Pascual (2005), for the first large-scale extension of the same design to a later sample; and Engel and West (2005), for the theoretical reconciliation of near-random-walk behavior with a genuine role for fundamentals.

13. Conclusion

ONE-SENTENCE THESIS

None of the leading structural or time-series models of exchange rate determination available in the early 1980s could out-forecast a driftless random walk at horizons of one to twelve months, even when supplied with the actual future values of their own explanatory variables, and this failure survives an unusually wide battery of robustness checks.

THREE MOST IMPORTANT INSIGHTS

First, in-sample fit is not a reliable guide to a model's forecasting value, and any empirical claim about exchange rate determination should be evaluated out-of-sample before it is trusted for policy or forecasting purposes. Second, the failure documented here is a property of the entire class of linear, fixed-coefficient models tested, including one (the VAR) with no exogeneity assumptions at all, which substantially narrows the plausible explanations to genuine misspecification, parameter instability, or a fundamentally efficient, unpredictable exchange rate process rather than to any single econometric technicality. Third, the paper's careful design — particularly its decision to supply realized future values of explanatory variables — means its negative result is unusually difficult to attribute to bad luck in forecasting the inputs, which is precisely what makes it durable.

STRONGEST EVIDENCE

The uniform ranking across all twelve exchange-rate/horizon cells in Table 3, holding under three different estimators, multiple monetary aggregates, multiple inflation proxies, and three different sample windows, together with the independent failure of the fully unrestricted vector autoregression.

BIGGEST WEAKNESS

The absence of a formal statistical test of the RMSE differences, a limitation the authors themselves flag and one that the Diebold–Mariano test, developed roughly a decade later, was specifically designed to remedy.

MOST IMPORTANT RESEARCH GAP

Whether the puzzle reflects genuine economic misspecification of the monetary approach or an undetected, unmodeled parameter instability coinciding with the 1970s oil shocks and the 1979 shift in U.S. monetary operating procedure — a question the original sample is simply too short to answer on its own.

BEST RESEARCH OPPORTUNITY

A rigorous Diebold–Mariano or Clark–West re-evaluation of the original and successor comparisons (Idea 1 above), because it directly addresses the paper's own most significant methodological limitation using only data that already exist.

THREE PAPERS TO READ NEXT

Dornbusch (1976), "Expectations and Exchange Rate Dynamics"; Cheung, Chinn, and Pascual (2005), "Empirical Exchange Rate Models of the Nineties: Are Any Fit to Survive?"; and Engel and West (2005), "Exchange Rates and Fundamentals."

14. References

Bilson, John F.O. (1978). Rational expectations and the exchange rate. In J. Frenkel and H. Johnson (eds.), The Economics of Exchange Rates. Addison-Wesley.

Bilson, John F.O. (1979). The Deutsche Mark/Dollar rate: a monetary analysis. In K. Brunner and A. Meltzer (eds.), Policies for Employment, Prices and Exchange Rates, Carnegie-Rochester Conference Series 11. North-Holland.

Bilson, John F.O. (1981). The speculative efficiency hypothesis. Journal of Business, 54, 435–451.

Calvo, Guillermo A. and Carmen M. Reinhart (2002). Fear of floating. Quarterly Journal of Economics, 117(2), 379–408.

Cheung, Yin-Wong, Menzie D. Chinn, and Antonio Garcia Pascual (2005). Empirical exchange rate models of the nineties: are any fit to survive? Journal of International Money and Finance, 24(7), 1150–1175.

Cheung, Yin-Wong, Menzie D. Chinn, Antonio Garcia Pascual, and Yi Zhang (2019). Exchange rate prediction redux: new models, new data, new currencies. Journal of International Money and Finance, 95, 332–362.

Clark, Todd E. and Kenneth D. West (2006). Using out-of-sample mean squared prediction errors to test the martingale difference hypothesis. Journal of Econometrics, 135(1–2), 155–186.

Diebold, Francis X. and Roberto S. Mariano (1995). Comparing predictive accuracy. Journal of Business and Economic Statistics, 13(3), 253–263.

Dornbusch, Rudiger (1976). Expectations and exchange rate dynamics. Journal of Political Economy, 84(6), 1161–1176.

Evans, Martin D.D. and Richard K. Lyons (2002). Order flow and exchange rate dynamics. Journal of Political Economy, 110(1), 170–180.

Fair, Ray C. (1970). The estimation of simultaneous equations models with lagged endogenous variables and first order serially correlated errors. Econometrica, 38(3), 507–516.

Frankel, Jeffrey A. (1979). On the mark: a theory of floating exchange rates based on real interest differentials. American Economic Review, 69(4), 610–622.

Frenkel, Jacob A. (1976). A monetary approach to the exchange rate: doctrinal aspects and empirical evidence. Scandinavian Journal of Economics, 78(2), 200–224.

Granger, Clive W.J. and Paul Newbold (1977). Forecasting Economic Time Series. Academic Press.

Haynes, Stephen E. and Joe A. Stone (1981). On the mark: comment. American Economic Review, 71(5), 1060–1067.

Hooper, Peter and John E. Morton (1982). Fluctuations in the dollar: a model of nominal and real exchange rate determination. Journal of International Money and Finance, 1, 39–56.

Meese, Richard A. and Kenneth Rogoff (1983). Empirical exchange rate models of the seventies: do they fit out of sample? Journal of International Economics, 14(1–2), 3–24.

Meese, Richard A. and Kenneth Rogoff (1983). The out-of-sample failure of empirical exchange rate models: sampling error or misspecification? In J. Frenkel (ed.), Exchange Rates and International Macroeconomics. University of Chicago Press.

Mendonça, Helder Ferreira de (2025). Fundamentals models versus random walk: evidence from an emerging economy. Journal of Forecasting, forthcoming.

Mussa, Michael (1979). Empirical regularities in the behavior of exchange rates and theories of the foreign exchange market. In K. Brunner and A. Meltzer (eds.), Policies for Employment, Prices and Exchange Rates, Carnegie-Rochester Conference Series 11. North-Holland.

Parzen, Emanuel (1975). Multiple time series: determining the order of approximating autoregressive schemes. Technical Report 23, State University of New York at Buffalo.

Rogoff, Kenneth (1996). The purchasing power parity puzzle. Journal of Economic Literature, 34(2), 647–668.

Simpson, Thomas D. and Richard D. Porter (1980). Some issues involving the definition and interpretation of the monetary aggregates. In Controlling Monetary Aggregates III, Federal Reserve Bank of Boston Conference Series 23.

Sims, Christopher A. (1980). Macroeconomics and reality. Econometrica, 48(1), 1–48.

Taylor, Mark P., David A. Peel, and Lucio Sarno (2001). Nonlinear mean-reversion in real exchange rates: toward a solution to the purchasing power parity puzzles. International Economic Review, 42(4), 1015–1042.

Working, Holbrook (1960). Note on the correlation of first differences of averages in a random chain. Econometrica, 28(4), 916–918.